

Problem 3 — Full Walkthrough

Stationary points of $x^2 + y^2 + \beta xy + x + 2y$

Goal: For each value of the scalar β , find every stationary point of

$$f(x, y) = x^2 + y^2 + \beta xy + x + 2y, \quad (x, y) \in \mathbb{R}^2,$$

and determine which ones (if any) are global minima.

Step 1 — Write f in matrix quadratic form

Let $v = (x, y)$, $M = \begin{pmatrix} 2 & \beta \\ \beta & 2 \end{pmatrix}$, $c = (1, 2)$. Claim: $f(v) = \frac{1}{2}v^T M v + c^T v$. Check by expanding:

$$\frac{1}{2}v^T M v = \frac{1}{2}(2x^2 + 2\beta xy + 2y^2) = x^2 + \beta xy + y^2, \quad c^T v = x + 2y.$$

Adding these gives $x^2 + \beta xy + y^2 + x + 2y$, which matches $f(x, y)$ exactly. So $f(v) = \frac{1}{2}v^T M v + c^T v$ — the same general quadratic pattern as an unconstrained quadratic optimization problem, with this particular M and c .

Step 2 — Gradient and the stationary-point system

For a quadratic function of this form, $\nabla f(v) = Mv + c$ (this can be checked directly, coordinate by coordinate):

$$\frac{\partial f}{\partial x} = 2x + \beta y + 1, \quad \frac{\partial f}{\partial y} = \beta x + 2y + 2.$$

Setting $\nabla f = 0$ gives $Mv = -c$, i.e. the linear system

$$2x + \beta y = -1, \quad \beta x + 2y = -2.$$

Step 3 — Solve the system (general β)

Eliminate y : multiply the first equation by 2 and the second by β :

$$4x + 2\beta y = -2, \quad \beta^2 x + 2\beta y = -2\beta.$$

Subtracting:

$$(4 - \beta^2)x = -2 - (-2\beta) = 2\beta - 2 \implies x^* = \frac{2(\beta - 1)}{4 - \beta^2}, \quad \text{provided } 4 - \beta^2 \neq 0.$$

Eliminate x similarly: multiply the first equation by β and the second by 2:

$$2\beta x + \beta^2 y = -\beta, \quad 2\beta x + 4y = -4.$$

Subtracting:

$$(\beta^2 - 4)y = -\beta - (-4) = 4 - \beta \implies y^* = \frac{4 - \beta}{\beta^2 - 4} = \frac{\beta - 4}{4 - \beta^2}.$$

So whenever $4 - \beta^2 \neq 0$ (i.e. $\beta \neq \pm 2$), there is a unique stationary point:

$$x^* = \frac{2(\beta - 1)}{4 - \beta^2}, \quad y^* = \frac{\beta - 4}{4 - \beta^2}.$$

Step 4 — Eigenvalues and eigenvectors of M

Test the vectors $v_1 = (1, 1)$ and $v_2 = (1, -1)$ directly:

$$Mv_1 = \begin{pmatrix} 2 + \beta \\ \beta + 2 \end{pmatrix} = (2 + \beta)v_1, \quad Mv_2 = \begin{pmatrix} 2 - \beta \\ \beta - 2 \end{pmatrix} = (2 - \beta)v_2.$$

So v_1, v_2 are eigenvectors of M with eigenvalues $2 + \beta$ and $2 - \beta$ respectively. Note $v_1 \cdot v_2 = 1 - 1 = 0$: they are orthogonal, and since $\mathbb{R}^2$ is 2-dimensional, $\{v_1, v_2\}$ is a complete orthogonal basis — these are *all* the eigenvalues of M , no others exist.

Step 5 — An explicit test for definiteness

Any vector $v \in \mathbb{R}^2$ can be written as $v = c_1 v_1 + c_2 v_2$. Since $Mv_1 = (2 + \beta)v_1$ and $Mv_2 = (2 - \beta)v_2$:

$$v^T M v = (c_1 v_1 + c_2 v_2)^T (c_1 (2 + \beta)v_1 + c_2 (2 - \beta)v_2).$$

Expanding, and using $v_1 \cdot v_2 = 0$, $v_1 \cdot v_1 = v_2 \cdot v_2 = 2$:

$$v^T M v = 2c_1^2(2 + \beta) + 2c_2^2(2 - \beta).$$

This single formula tells us everything about M 's definiteness as β varies:

- If $-2 < \beta < 2$: both $(2 + \beta) > 0$ and $(2 - \beta) > 0$, so $v^T M v > 0$ for every nonzero v (some c_1 or c_2 must be nonzero). **M is positive definite.**
- If $\beta = 2$: $v^T M v = 4c_1^2 \geq 0$, with equality whenever $c_1 = 0$ (i.e. v is any multiple of v_2) — **M is positive semidefinite, not definite.**
- If $\beta = -2$: symmetric situation, $v^T M v = 4c_2^2 \geq 0$, equality when v is a multiple of v_1 — **positive semidefinite, not definite.**
- If $|\beta| > 2$: $(2 + \beta)$ and $(2 - \beta)$ have opposite signs, so $v^T M v$ is positive for $v = v_1$ alone and negative for $v = v_2$ alone — **M is indefinite.**

Step 6 — Case analysis

Case $|\beta| < 2$. From Step 3, the unique stationary point (x^*, y^*) exists. From Step 5, $M \succ 0$ everywhere (it's constant, so this holds globally, not just at (x^*, y^*)) — f is strictly convex. A stationary point of a strictly convex function is its unique global minimum. **(x^*, y^*) is the unique global minimum.**

Case $\beta = 2$. The system from Step 2 becomes

$$2x + 2y = -1, \quad 2x + 2y = -2.$$

Subtracting the two equations gives $0 = 1$ — a contradiction. **No stationary point exists.** To see f is unbounded below, move along the direction $v_2 = (1, -1)$ (the null eigenvector) starting from the origin: $v = (t, -t)$. Using $f(v) = \frac{1}{2}v^T M v + c^T v$ and Step 5's formula with $c_1 = 0, c_2 = t$ (since $(t, -t) = t \cdot v_2$):

$$\frac{1}{2}v^T M v = \frac{1}{2} \cdot 2t^2(2 - \beta) = t^2(2 - 2) = 0, \quad c^T v = (1, 2) \cdot (t, -t) = t - 2t = -t.$$

So $f(t, -t) = 0 + (-t) = -t \rightarrow -\infty$ as $t \rightarrow \infty$. **Unbounded below.**

Case $\beta = -2$. The system becomes

$$2x - 2y = -1, \quad -2x + 2y = -2.$$

Adding the two equations gives $0 = -3$ — a contradiction. **No stationary point exists.** Move along $v_1 = (1, 1)$: $v = (t, t)$, so $c_1 = t, c_2 = 0$:

$$\frac{1}{2}v^T M v = \frac{1}{2} \cdot 2t^2(2 + \beta) = t^2(2 - 2) = 0, \quad c^T v = (1, 2) \cdot (t, t) = 3t.$$

So $f(t, t) = 3t \rightarrow -\infty$ as $t \rightarrow -\infty$. **Unbounded below.**

Case $|\beta| > 2$. From Step 3, the unique stationary point (x^*, y^*) still exists (since $4 - \beta^2 \neq 0$). From Step 5, M is indefinite: (x^*, y^*) is a **saddle point**, not a local minimum.

To confirm f is unbounded below, use the exact quadratic expansion around the stationary point. Since f is quadratic, its Taylor expansion around any point is exact with no remainder:

$$f(v^* + w) = f(v^*) + \nabla f(v^*)^T w + \frac{1}{2}w^T M w.$$

At the stationary point, $\nabla f(v^*) = 0$, so the linear term vanishes entirely:

$$f(v^* + w) = f(v^*) + \frac{1}{2}w^T M w.$$

Take $w = t \cdot v_2 = (t, -t)$, the eigenvector for the *negative* eigenvalue when $\beta > 2$ (since $2 - \beta < 0$ there):

$$f(v^* + tv_2) = f(v^*) + \frac{1}{2} \cdot 2t^2(2 - \beta) = f(v^*) + t^2(2 - \beta) \rightarrow -\infty \text{ as } t \rightarrow \infty,$$

since $(2 - \beta) < 0$. (For $\beta < -2$, use $w = t \cdot v_1$ instead, since then $2 + \beta < 0$ is the negative eigenvalue; the same computation gives $f(v^* + tv_1) = f(v^*) + t^2(2 + \beta) \rightarrow -\infty$.) **Unbounded below; no global minimum exists.**

Summary

Range of β	Stationary points	Classification
$-2 < \beta < 2$	unique: (x^*, y^*) from Step 3	unique global minimum
$\beta = 2$	none	unbounded below
$\beta = -2$	none	unbounded below
$ \beta > 2$	unique: (x^*, y^*) from Step 3	saddle point; unbounded below

A global minimum exists — and is the unique stationary point — exactly when $|\beta| < 2$. In every other case, $\inf f = -\infty$.